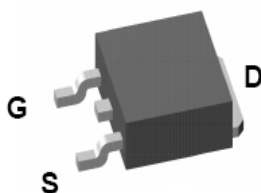


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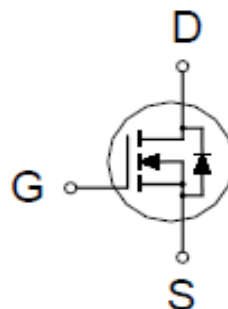
N-Channel Enhancement Mode MOSFET

PRODUCT SUMMARY

$V_{(BR)DSS}$	$R_{DS(ON)}$	I_D
25V	9.5mΩ @ $V_{GS} = 10V$	56A



TO-252



ABSOLUTE MAXIMUM RATINGS ($T_C = 25\text{ }^{\circ}\text{C}$ Unless Otherwise Noted)

PARAMETERS/TEST CONDITIONS		SYMBOL	LIMITS	UNITS
Drain-Source Voltage		V_{DS}	25	V
Gate-Source Voltage		V_{GS}	±20	
Continuous Drain Current	$T_C = 25\text{ }^{\circ}\text{C}$	I_D	56	A
	$T_C = 100\text{ }^{\circ}\text{C}$		35	
Pulsed Drain Current ¹		I_{DM}	160	
Avalanche Current		I_{AS}	34	
Avalanche Energy	$L=0.1\text{mH}$	E_{AS}	60	mJ
Power Dissipation	$T_C = 25\text{ }^{\circ}\text{C}$	P_D	49	W
	$T_C = 100\text{ }^{\circ}\text{C}$		20	
Junction & Storage Temperature Range		T_J, T_{stg}	-55 to 150	$^{\circ}\text{C}$
Lead Temperature (1/16" from case for 10 sec.)		T_L	275	

THERMAL RESISTANCE RATINGS

THERMAL RESISTANCE	SYMBOL	TYPICAL	MAXIMUM	UNITS
Junction-to-Case	$R_{\theta JC}$		2.55	$^{\circ}\text{C} / \text{W}$
Junction-to-Ambient	$R_{\theta JA}$		63	

¹Pulse width limited by maximum junction temperature.

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ELECTRICAL CHARACTERISTICS ($T_C = 25^\circ\text{C}$, Unless Otherwise Noted)

PARAMETER	SYMBOL	TEST CONDITIONS	LIMITS			UNITS
			MIN	TYP	MAX	
STATIC						
Drain-Source Breakdown Voltage	$V_{(BR)DSS}$	$V_{GS} = 0V, I_D = 250\mu A$	25			V
Gate Threshold Voltage	$V_{GS(th)}$	$V_{DS} = V_{GS}, I_D = 250\mu A$	1	1.6	3	
Gate-Body Leakage	I_{GSS}	$V_{DS} = 0V, V_{GS} = \pm 25V$			± 100	nA
Zero Gate Voltage Drain Current	I_{DSS}	$V_{DS} = 20V, V_{GS} = 0V$			1	μA
		$V_{DS} = 20V, V_{GS} = 0V, T_J = 125^{\circ}C$			10	
Drain-Source On-State Resistance ¹	$R_{DS(ON)}$	$V_{GS} = 5V, I_D = 20A$		12	19	m Ω
		$V_{GS} = 10V, I_D = 25A$		7	9.5	
Forward Transconductance ¹	g_{fs}	$V_{DS} = 15V, I_D = 20A$		60		S
DYNAMIC						
Input Capacitance	C_{iss}	$V_{GS} = 0V, V_{DS} = 15V, f = 1MHz$		1400		pF
Output Capacitance	C_{oss}			300		
Reverse Transfer Capacitance	C_{rss}			190		
Gate Resistance	R_g	$V_{GS} = 0V, V_{DS} = 0V, f = 1MHz$		1.3		Ω
Total Gate Charge ²	Q_g	$V_{GS} = 10V$	$V_{DS} = 0.5V_{(BR)DSS},$ $I_D = 25A$	25		nC
		$V_{GS} = 4.5V$		11		
Gate-Source Charge ²	Q_{gs}	6				
Gate-Drain Charge ²	Q_{gd}	5				
Turn-On Delay Time ²	$t_{d(on)}$	$V_{DS} = 15V, R_L = 15\Omega$ $I_D \cong 1A, V_{GS} = 10V, R_{GEN} = 6\Omega$		16		
Rise Time ²	t_r		25			
Turn-Off Delay Time ²	$t_{d(off)}$		60			
Fall Time ²	t_f		16			
SOURCE-DRAIN DIODE RATINGS AND CHARACTERISTICS ($T_C = 25^{\circ}C$)						
Continuous Current	I_S				37	A
Forward Voltage ¹	V_{SD}	$I_F = I_S, V_{GS} = 0V$			1.3	V
Reverse Recovery Time	t_{rr}	$I_F = 25A, dI_F/dt = 100A/\mu S$		35		nS
Reverse Recovery Charge	Q_{rr}			61		nC

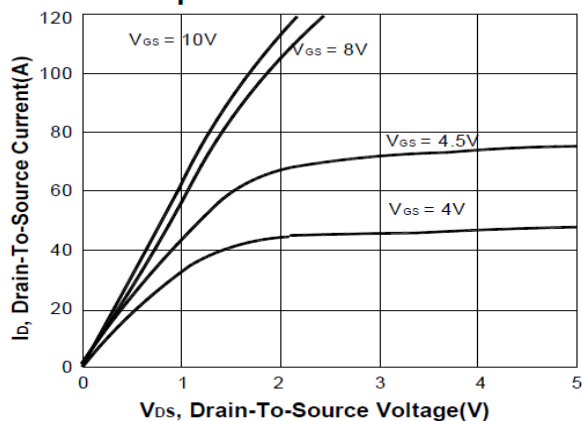
¹Pulse test : Pulse Width $\leq 300 \mu\text{sec}$, Duty Cycle $\leq 2\%$.

²Independent of operating temperature.

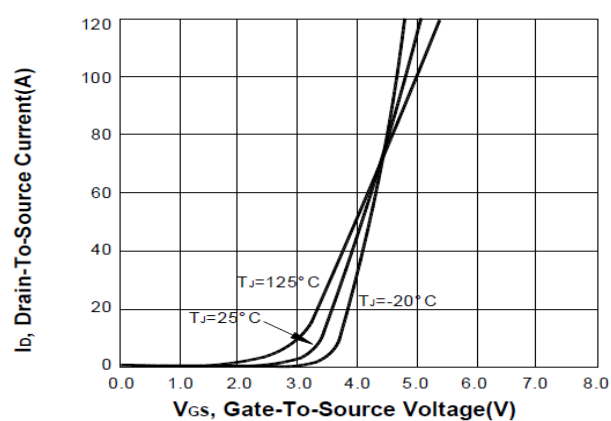
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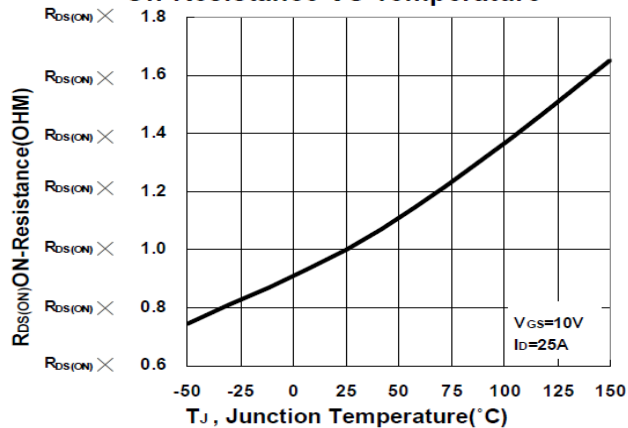
Output Characteristics



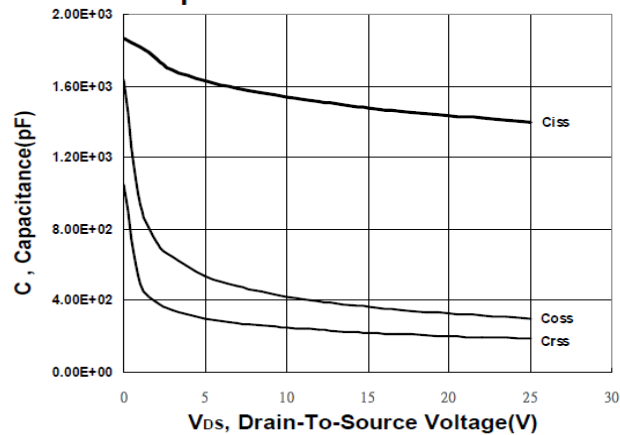
Transfer Characteristics



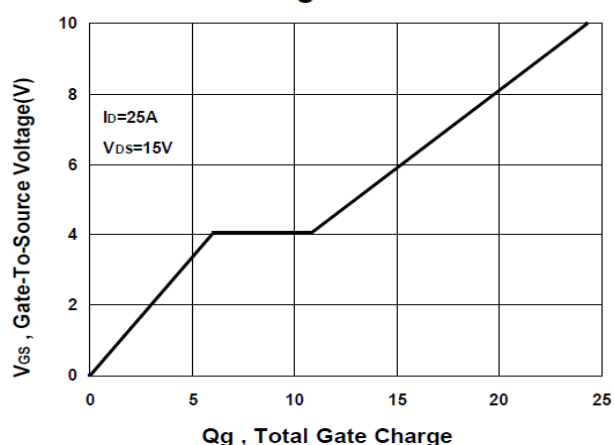
On-Resistance VS Temperature



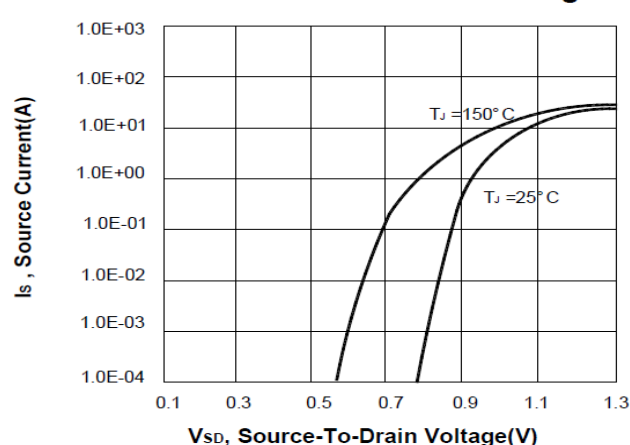
Capacitance Characteristic



Gate charge Characteristics



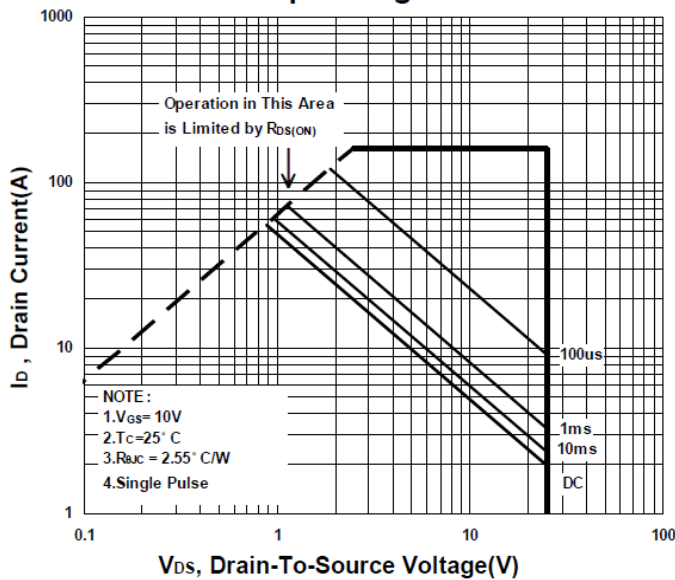
Source-Drain Diode Forward Voltage



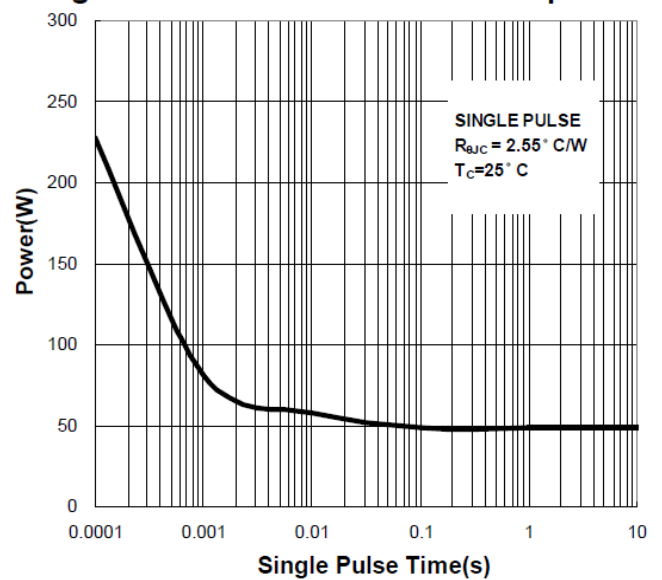
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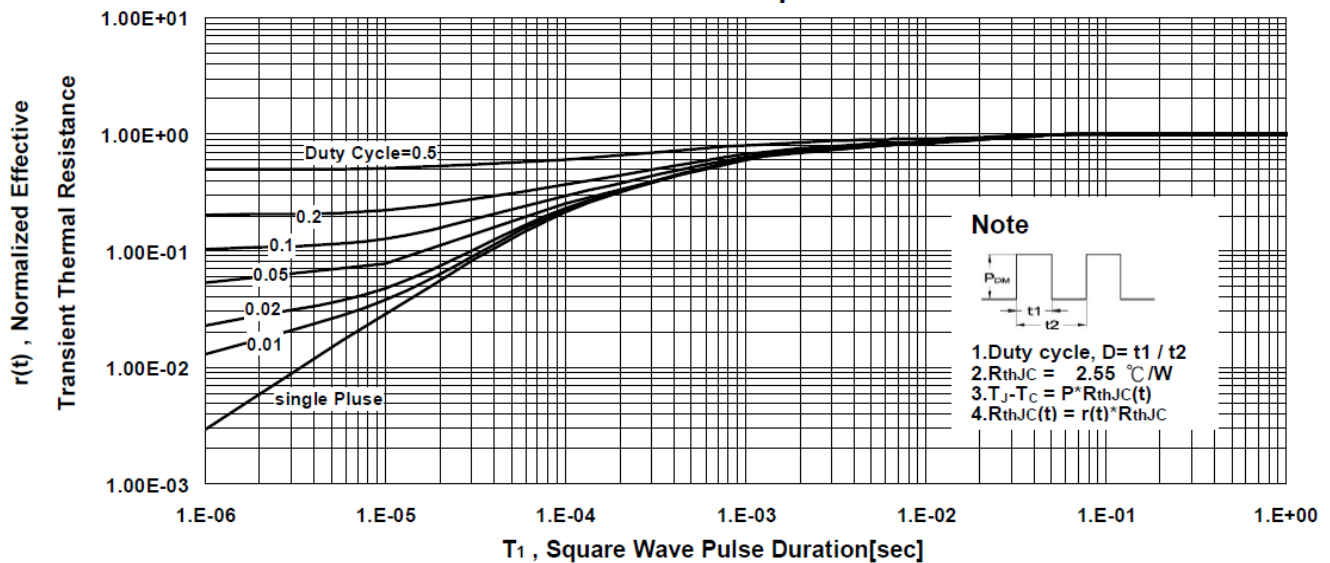
Safe Operating Area



Single Pulse Maximum Power Dissipation



Transient Thermal Response Curve



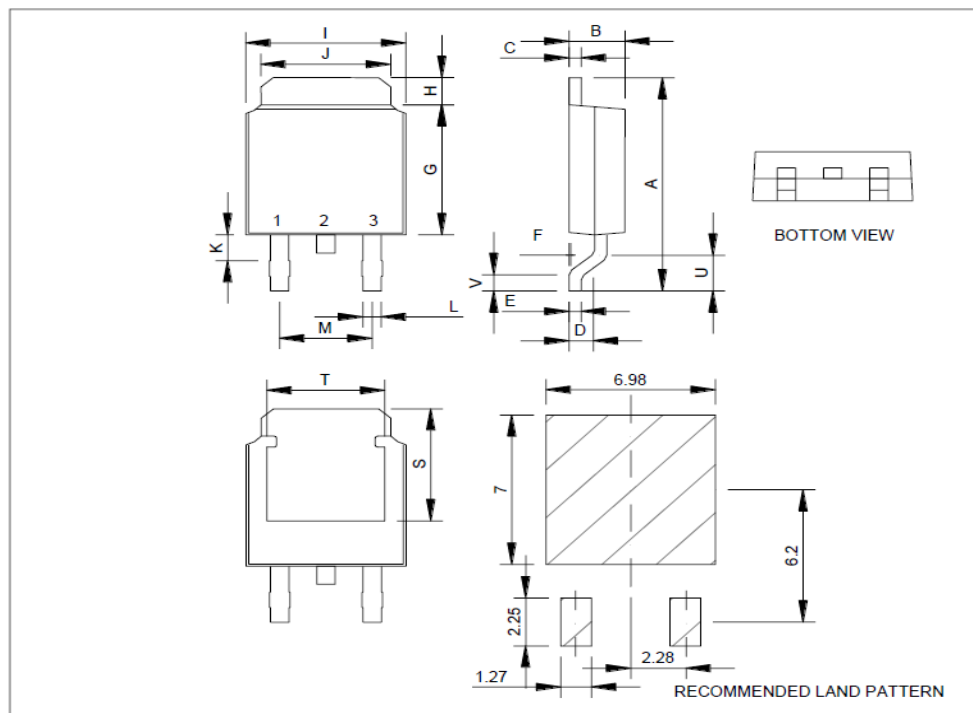
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Package Dimension

TO-252 (DPAK) MECHANICAL DATA

Dimension	mm			Dimension	mm		
	Min.	Typ.	Max.		Min.	Typ.	Max.
A	8.9	10	10.41	J	4.8		5.64
B	2.1	2.2	2.4	K	0.15		1.1
C	0.4	0.5	0.61	L	0.4	0.76	0.89
D	0.82	1.2	1.5	M	4.2	4.58	5
E	0.4	0.5	0.61	S	4.9	5.1	5.3
F	0		0.2	T	4.6	4.75	5.44
G	5.3	6.1	6.3	U	1.4		1.78
H	0.9		1.7	V	0.55	1.25	1.7
I	6.3	6.5	6.8				



*因为各家封装模具不同而外观略有所差异，不影响电性及Layout。